

GRIDGUARD AI

Power Outage Prediction & Grid Equipment Failure Advisor

Transforming electric utility operations from reactive catastrophe response into proactive, automated resilience by fusing multi-modal sensor telemetry, spatial weather forecasts, and historical failure records with IBM Bob and watsonx.ai.

PROJECT TEAM: GridGuard AI

- Harshil Thakkar (Lead) — Full-Stack & System Architect
- Priyal Rathod — AI/ML Risk Modeling & Physics Engine
- Rutvik Jasani — Backend & SCADA Telemetry Gateway
- Hetavi Suhagiya — Frontend UI & Data Visualization

CORE TECHNOLOGIES

- IBM Bob CLI & Model Context Protocol (MCP)
- IBM watsonx.ai (Granite / Llama-3 Foundation Models)
- Python FastAPI, NumPy, IEEE C57 / IEC 60076 Standards
- React 18, TypeScript, Recharts, SCADA HUD Console

Aging Power Grids Are Blind to Compounding Weather Failures

Traditional calendar maintenance leaves multi-million-dollar transformers vulnerable to unpredicted catastrophic breakdowns.

SCADA GRID DISPATCHERS

- Overwhelmed by 500+ alarms/hr during storms
- Current SCADA systems only alert AFTER equipment trips
- Zero forward visibility into 12-72h degradation curves
- Forced to make high-stress decisions without physics root-cause explanations

FIELD CREW DIRECTORS

- Reliant on calendar schedules (every 12-24 mos)
- Internal insulation decay escalates between visits
- Emergency crews dispatched into impassable flooded roads after failure occurs
- High overtime costs and spare-parts logistics chaos

CITIZENS & CRITICAL ASSETS

- Blackouts sever power to hospitals, water pumps, and telecommunications
- Protracted outages lasting days to weeks
- Severe economic devastation to regional industry
- Public Utility Commission regulatory penalties

\$1M - \$2.5M

PER HOUR COST

Direct financial cost of major substation blackout in industrial corridors.

4 - 18 MONTHS

REPLACEMENT LEAD TIME

Global procurement delay to engineer & deliver a Large Power Transformer.

70% BUDGET

REACTIVE DRAIN

Utility maintenance expenditure consumed by emergency spot repairs.

500+ ALARMS

ALARM FLOOD / HR

Nuisance false-positive alarm triggers disorienting control room operators.

GridGuard AI: Closed-Loop Predictive Grid Resilience

Synthesizing IoT telemetry, weather forecasts, and historical logs into automated crew pre-positioning before outages strike.

WHAT WE BUILT

GridGuard AI transforms electric utility operations from reactive catastrophe response into proactive, automated resilience.

- **Physics-Informed Anomaly Engine:** Evaluates equipment health against IEEE C57.104 and IEC 60076 standards.
- **Dynamic Weather Risk Multiplier:** Correlates incoming atmospheric heatwaves and convective storm vectors with live thermal thresholds.
- **Downstream Criticality Prioritization:** Ranks substations by customer impact and critical infrastructure dependencies.

01 // MULTI-MODAL DATA FUSION

Ingests 4 IoT telemetry streams (Temperature, Vibration, Partial Discharge, DGA Oil Quality) + forward spatial weather models + historical maintenance dossiers.

02 // PHYSICS-GROUNDED FAILURE MODEL

Computes composite risk: $0.35 * \text{Sensor} + 0.30 * \text{Weather} + 0.20 * \text{History} + 0.15 * \text{Age}$.
Estimates remaining safe operating hours before thermal/dielectric runaway.

03 // GRID IMPACT SEVERITY RANKING

Scales failure probability by downstream customer density and hospital/critical node weighting:
 $\text{Severity} = \text{Probability} * \text{Grid Impact} * \text{Weather Factor}$.

04 // PRESCRIPTIVE MAINTENANCE & CREW STAGING

Generates concrete engineering work orders (DGA degassing, bushing replacement) and automatically stages emergency strike teams in advance.

High-Throughput, Resilient Multi-Tier Infrastructure

Stateless analytical microservices coupled with sub-second SCADA HUD telemetry rendering.

TIER 1: INGESTION

Data Fusion Engine

`data_fusion.py`

- Real-time IoT sensor bus parser
- 48h spatial weather API vectorizer
- Historical MTBF incident cross-ref
- Multi-source feature harmonization

TIER 2: AI ANALYTICS

FastAPI AI Engine

`Port 8001 / REST`

- IEEE C57 & IEC 60076 thresholding
- Vectorized multi-factor risk scoring
- Downstream severity ranking
- Sub-50ms inference latency

TIER 3: PERSISTENCE

Backend API Gateway

`Port 5000 / REST`

- Node.js Express 4 REST routing
- PostgreSQL & TimescaleDB hypertables
- Resilient offline fallback mock engine
- NERC-CIP audit trail buffering

TIER 4: SCADA CONSOLE

React 18 + Vite HUD

`Port 3000 / Web`

- 60 FPS HTML5 canvas electric arcs
- Multi-sensor Recharts trend lines
- 12 substation geospatial grid map
- Level 1/2/3 clearance studio

KEY ARCHITECTURAL HIGHLIGHTS

- ⚡ Zero-Downtime Evaluation: Built-in resilient fallback engine ensures full 12-substation interactivity even if local PostgreSQL is offline.
- 🔒 NERC-CIP Compliance: Role-based operational clearance (L1 Monitor, L2 Dispatch, L3 Chief), simulated hardware 2FA, and out-of-band OTP recovery.
- 🚀 Cloud Ready: Stateless microservices architecture deployable onto IBM Cloud Code Engine or Red Hat OpenShift with horizontal auto-scaling.

Operational Command Center & Asset Diagnostics

Real-time visibility across 12 high-voltage transmission nodes with interactive telemetry and crew dispatch.



COMMON GRID DASHBOARD

12 High-Voltage Nodes

Geospatial SCADA map color-coded by real-time risk severity. Criticality table automatically sorts substations by failure probability and downline customer impact.

ASSET DIGITAL TWIN & TRENDS

Multi-Sensor Recharts

Small-multiple time-series plots for Top-Oil Temperature, Vibration, Partial Discharge, and DGA Oil Quality. Pinpoints critical failure window (< 14 hours).

OPERATOR PROFILE & ENCLAVE

Clearance Studio (L1 / L2 / L3)

Role switcher for live hackathon demos. Rotates security keys, verifies out-of-band email OTP reset codes, and records an immutable NERC-CIP audit trail.

Where and How IBM Bob & watsonx Are Integrated

Combining IBM Bob conversational orchestration with watsonx.ai foundation models via Model Context Protocol.

IBM BOB CLI & AGENT

Conversational SCADA Advisor

Acts as the 24/7 intelligent control room co-pilot. Grid dispatchers interrogate complex anomalies using natural language queries ("Why is Substation Alpha at critical risk?") and receive instant, context-aware engineering diagnoses.

MODEL CONTEXT PROTOCOL (MCP)

Standardized SCADA Tool Bridge

Connects IBM Bob directly to live GridGuard backend REST endpoints. Exposes real-time tools to fetch asset telemetry, query weather risk multipliers, compute IEEE failure probabilities, and trigger emergency crew dispatch.

WATSONX.AI FOUNDATION MODELS

Enterprise Playbook Generation

Leverages IBM Granite and Llama-3 models to synthesize multi-sensor telemetry into structured, IEEE-compliant mitigation playbooks, load-shedding schedules, and regulatory NERC-CIP incident reports.

IBM CLOUD CODE ENGINE

Serverless Elastic Scaling

Runs containerized AI inference microservices on managed serverless infrastructure. Auto-scales computing instances instantaneously when severe weather systems compound grid-wide sensor ingestion loads.

[DISPATCHER PROMPT] → [IBM BOB AGENT] → [MCP TOOL CALL] → [SCADA TELEMETRY + WATSONX.AI] → [ACTIONABLE MITIGATION PLAYBOOK]

Projected Business Impact & Beyond the Hackathon

Delivering massive operational savings and a clear trajectory toward production utility deployment.

42% REDUCTION

UNPLANNED OUTAGE DURATION

Compresses utility SAIDI/SAIFI outage indices via predictive strike crew staging.

\$85M / YEAR

REGIONAL COST AVOIDANCE

Savings for a standard 2M-meter utility by eliminating catastrophic transformer replacements.

12,000+ HOURS

FIELD LABOR OPTIMIZATION

Eliminates wasteful calendar-based manual inspections with continuous condition monitoring.

PHASE 1 (Q3-Q4 2026)

Physical SCADA & Hardware Integration

- Connect to physical substation RTUs via DNP3 & IEC 61850 protocols.
- Deploy edge inference gateways onto substation control house hardware.
- Ingest live optical fiber partial discharge & continuous online DGA monitors.

PHASE 2 (Q1-Q2 2027)

Autonomous Drone & Robotics Dispatch

- Automated drone inspection launch triggers for thermal infrared substation scans.
- Real-time road flood & downed-tree satellite routing for pre-positioned fleets.
- Closed-loop automated breaker tripping recommendations.

PHASE 3 (2027+)

Regional RTO Federated Intelligence

- Multi-utility threat intelligence sharing across ISO / RTO interconnections.
- Cross-border cascading blackout mitigation with watsonx distributed models.
- Commercial SaaS deployment across North American & European utilities.

Engineering the Future of Resilient Energy Grids

A complete, production-grade AI platform built for the IBM Bob AI Hackathon.

Harshil Thakkar

TEAM LEAD & FULL-STACK
ARCHITECT

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- Overall platform architecture
- React SCADA HUD & canvas
- FastApi / Express integration
- End-to-end evaluation pipeline

Priyal Rathod

AI/ML RISK ENGINEER

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- IEEE C57 / IEC 60076 scoring
- Multi-modal data fusion pipeline
- Weather compounding model
- Time-to-failure forecasting

Rutvik Jasani

BACKEND & SCADA GATEWAY

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- Node.js REST API gateway
- PostgreSQL / TimescaleDB schema
- Resilient fallback engine
- NERC-CIP audit logging

Hetavi Suhagiya

FRONTEND & DATA VIZ

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- Recharts sensor telemetry
- Geospatial substation map
- Operator clearance studio
- UI design tokens & accessibility

⚡ SUBMISSION ARTIFACTS & LINKS

- Source Code: github.com/harshil5506/bob-ai-hackathon-GridGaurdAI
- Comprehensive Documentation: [docs/problem-statement.md](#) | [docs/solution-overview.md](#) | [docs/architecture.md](#) | [docs/setup-guide.md](#)
- Live SCADA Console: <http://localhost:3000> (Pre-loaded 1-Click Chief Controller Login)
- Mission: Protecting the critical electric power that powers modern civilization.